

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 07 · LECTURE NOTES

Alignment and Stitching

Putting several photographs into one coordinate system.

MODULE OUTCOMES

- Explain when a homography is the correct motion model.
- Implement feature detection, description, and matching.
- Implement RANSAC and explain why robust estimation is necessary.
- Explain bundle adjustment.
- Implement seamless blending.

1 Motion models

Model	DOF	Preserves	Correct when
Translation	2	Everything but position.	Small shifts — burst alignment (Module 10), where frames differ by hand tremor.
Euclidean	3	Lengths and angles.	Rotation about the optical axis plus translation.
Similarity	4	Angles and length ratios.	The above plus uniform scale.
Affine	6	Parallelism.	Weak perspective — a distant planar surface, or a long lens.
Homography (projective)	8	Straight lines, and nothing else.	A camera rotating about its optical centre, or a planar scene. Those two cases and no others.

PANORAMAS REQUIRE ROTATION ABOUT THE OPTICAL CENTRE

A homography can relate two views only when there is no parallax. That holds in exactly two circumstances: the camera rotated about its optical centre without translating, or everything photographed lies in a single plane.

Translate the camera and nearby objects shift relative to distant ones. That relative shift is parallax, it carries depth information, and **no single 2D transform can account for it** — aligning the background necessarily misaligns the foreground.

Which is why handheld panoramas fail on close subjects. Rotating your body moves the camera by tens of centimetres, so a foreground object at two metres and a background at fifty cannot both be aligned. Distant landscapes stitch beautifully; a room does not.

And it is why panoramic tripod heads rotate about the lens's nodal point rather than about the tripod screw. The difference is a few centimetres of offset, and it is the difference between a panorama that works and one that cannot be made to.

2 Finding correspondences

- 1 **Detect.** Find points that can be located repeatably in both images and that are distinctive enough to match. **Corners, not edges** — a point on an edge cannot be localised along the edge's direction, which is the **aperture problem**, and it makes edge points useless as correspondences.
- 2 **Describe.** Summarise each point's neighbourhood as a vector that is invariant to whatever you expect to change between the images — rotation, scale, and illumination, typically.
- 3 **Match.** For each descriptor in one image, find the nearest in the other, and keep the plausible pairs.
- 4 **Apply the ratio test.** Accept a match only if the nearest neighbour is substantially closer than the second nearest — Lowe suggested a ratio of 0.8. **A point with two equally good matches has no reliable match at all**, which is exactly the situation produced by repeated texture. **This is one line of code and it removes the large majority of bad matches**, and it is Lowe's most useful practical contribution.

Detector / descriptor	Mechanism	Assessment
Harris corner	Eigenvalues of the local structure tensor — both large means a corner.	Classic, cheap, and not scale invariant , so it fails when the images differ in zoom.
SIFT	Scale-space extrema of a difference-of-Gaussian pyramid, with a descriptor built from histograms of gradient orientation in a 4×4 grid.	The standard for two decades. Scale and rotation invariant, robust to illumination because the descriptor is normalised. Patent expired in 2020 , so it is now freely usable.
SURF	An approximation of SIFT using box filters and integral images.	Faster, slightly less accurate. Largely superseded.

Detector / descriptor	Mechanism	Assessment
ORB	FAST corner detection with a BRIEF binary descriptor, plus rotation handling.	Very fast, free, and adequate for most alignment tasks. Binary descriptors compare with Hamming distance, which is extremely cheap. The practical default.
Learned detectors and descriptors	Trained end to end on correspondence tasks.	Better on hard cases — large viewpoint change, repeated structure — at greater cost.

MOST MATCHES ARE WRONG, WHICH IS WHY RANSAC EXISTS

Even after the ratio test, a substantial fraction of the surviving matches are incorrect. Repeated texture, moving objects, and genuine ambiguity all produce confident wrong answers.

Least-squares fitting cannot tolerate this. The squared error of an outlier grows without bound, so a single gross mismatch can pull the estimated transform arbitrarily far from the truth. **One bad match out of a hundred can ruin the fit.**

RANSAC — random sample consensus — inverts the approach: repeatedly draw a *minimal* set of correspondences, fit a transform to just those, and count how many of the remaining matches agree with it. Keep the hypothesis with the most support, then refit using all of its inliers.

Four point correspondences determine a homography, so the minimal sample is small — and the probability that four randomly chosen matches are all correct is workable even when half the matches are wrong, which is what makes the method practical. The required iteration count follows from the inlier ratio and can be computed rather than guessed.

3 Global consistency and blending

PAIRWISE ALIGNMENT DRIFTS; BUNDLE ADJUSTMENT FIXES IT

Chaining pairwise estimates accumulates error. Align image 1 to 2, then 2 to 3, then 3 to 4, and each small error compounds into the next. By the tenth image the misalignment is visible.

And a full 360-degree panorama does not close. Coming all the way round, the last image does not meet the first — the accumulated error appears as a visible discontinuity, and simply forcing the ends together distributes the error badly.

Bundle adjustment solves for all camera parameters simultaneously, minimising the total reprojection error across every correspondence in every image pair at once. The constraint that the panorama closes is then satisfied naturally rather than imposed.

It is a large sparse non-linear least-squares problem, solved by Levenberg–Marquardt exploiting the sparsity structure. **And it is the same machinery underlying structure from motion and SLAM** — including the inside-out tracking of CSCE 650 Module 05, which is bundle adjustment running continuously.

- **Exposure differs between frames.** Vignetting darkens each image's corners, automatic exposure changes between shots, and the light itself may change. **A hard seam is therefore always visible**, even with perfect geometric alignment.
- **Feathering** linearly blends across the overlap region. Simple, and **it ghosts whenever alignment is imperfect**, because it averages two slightly different images.
- **Multi-band blending** decomposes both images into Laplacian pyramids (Module 02) and blends each frequency band over a different width — low frequencies over a wide region to hide exposure differences, high frequencies over a narrow one to avoid ghosting detail. **This is the standard answer**, and it is why the Laplacian pyramid's invertibility matters.
- **Graph cut seam finding** chooses *where* to put the seam, by finding a path through the overlap along which the two images already agree — routing around a moving object rather than averaging it.
- **Gradient-domain blending** (Module 08) is the most powerful approach and the subject of the next module.

CHOOSE THE SEAM, THEN BLEND ACROSS IT

Blending can hide a small exposure difference. It cannot hide a person who moved between frames — averaging two images in which someone is in different places produces a ghost, and no amount of feathering removes it.

Graph cut seam selection finds a cut through the overlap region where the two images already agree, treating disagreement as cost. The seam routes around the moving person, taking them entirely from one image or the other.

Then multi-band blending across that chosen seam removes the residual exposure and vignetting difference, which the seam selection does not address.

Seam selection plus blending is what production stitchers do, and either technique alone is noticeably worse: blending alone ghosts moving objects, and seam selection alone leaves a visible exposure step.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Szeliski — Computer Vision, Chapters 8 and 9 (free)	Alignment, motion models, RANSAC, bundle adjustment, and stitching. The reference for this module , and Szeliski wrote the standard panorama survey as well.
Lowe — Distinctive Image Features from Scale-Invariant Keypoints (free)	SIFT, including the ratio test of §2. One of the most cited papers in computer vision, and readable.
Brown & Lowe — Automatic Panoramic Image Stitching (free)	The complete pipeline of this module, end to end, including bundle adjustment and multi-band blending.
Burt & Adelson — A Multiresolution Spline with Application to Image Mosaics (free)	Multi-band blending, from 1983 and still the method in use.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Photograph a scene with a close foreground and a distant background, rotating your body between shots. Attempt to stitch and demonstrate the parallax failure.
- 2 Repeat, rotating about the lens's approximate nodal point, and show the improvement.
- 3 Implement Harris corner detection and show it fails under scale change.

- 4 Implement or apply SIFT or ORB matching. Visualise the matches before and after the ratio test and count how many it removed.
- 5 Implement RANSAC for homography estimation. Plot the estimated transform error against the fraction of outliers, for least squares and for RANSAC.
- 6 Compute the number of RANSAC iterations needed for a given inlier ratio and confidence, and verify it empirically.
- 7 Stitch five images by chaining pairwise homographies and measure the drift at the far end.
- 8 Implement feathering and multi-band blending on the same overlap. Photograph something with fine texture to make the ghosting visible.
- 9 Stitch a panorama containing a person who moved. Implement graph cut seam selection and show the ghost removed.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name five motion models with their degrees of freedom, and say when each is correct.
- 2 Under exactly what two conditions is a homography exact?
- 3 Why do handheld panoramas fail on close subjects?
- 4 Why detect corners rather than edges?
- 5 What is the ratio test and why does it work?
- 6 Why can least squares not be used on raw feature matches?
- 7 Describe RANSAC and say why four points matter for a homography.
- 8 Why does pairwise alignment drift, and what does bundle adjustment do?
- 9 Name four blending approaches and say which two should be combined.